

CO2 – DEBUGGING OUTPUT

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Course Code: CSA0311

Course Name: Data Structure

Question 1

Buggy code in node creation

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

int main() {
    struct Node *newNode;
    newNode->data = 10;
    newNode->next = NULL;
    printf("&quot;%d&quot;, newNode->data);
    return 0;
}
```

OUTPUT:

```
//
int main() {
    struct Node *newNode;
    newNode->data = 10;
    newNode->next = NULL;
    printf("%d", newNode->data);
    return 0;
}
```

Your Code

```
1 #include<stdio.h>
2
3 struct Node{
4     int data;
5     struct Node *next;
6 };
7 int main() {
8     struct Node *newNode;
9     newNode = malloc(sizeof(struct Node));
10    newNode->data = 10;
11    newNode->next = NULL;
12    printf("%d", newNode->data);
13    free(newNode);
14    return 0;
15 }
```

Input

stdin input

Output

10

Question 2

Include missing statements and display the list of elements in linkedlist

```
void display(struct Node *head)
```

```
{
```

```
while(head != NULL);
```

```
{
```

```
printf("&quot;%d &quot;, head-&gt;data);
```

```
head = head-&gt;next;
```

```
} }
```

Your Code

```
1 #include<stdio.h>
2
3 struct Node{
4     int data;
5     struct Node*next;
6 };
7 void display(struct Node *head){
8     while(head != NULL){
9         printf("%d",head -> data);
10        head = head-> next;
11    }
12 }
13 int main() {
14     struct Node *head,*second,*third;
15     head =(struct Node*)malloc (sizeof(struct Node));
16     second = (struct Node*)malloc (sizeof(struct Node));
17     third = (struct Node*)malloc(sizeof(struct Node));
18     head -> data = 10;
19     second -> data = 20;
20     third -> data = 30;
21     head->next = second;
22     second->next = third;
23     third->next = NULL;
24     printf("linkes list: ");
25     display(head);
26     free(head);
```

Input

stdin input

Output

linkes list: 102030

Question 3

Find the bugs, include the missing lines in the following snippet.

```
int linearSearch(int arr[], int n, int key) {  
    for(int i = 0; i &lt;= n; i++) {  
        if(arr[i] == key)  
            return i;  
    }  
    return -1;  
}
```

```
}
```

Your Code

```
1 #include<stdio.h>  
2 int linearSearch(int arr[],int n,int key){  
3     for(int i=0;i<n;i++){  
4         if(arr[i]==key){  
5             return i;  
6         }  
7     }  
8     return -1;  
9 }  
10 int main() {  
11     int arr[]={10,20,30,40,50};  
12     int n=5;  
13     int key=30;  
14     int result=linearSearch(arr,n,key);  
15     if(result!=-1)  
16         printf("Element found at index %d\n",result);  
17     else  
18         printf("Element not found\n");  
19     return 0;  
20 }
```

Input

stdin input

Output

Element found at index 2